

PRALAY
Short-range, surface-to-surface missile ideal for strikes on targets like command centres and airbases
Range: **150-500 km**
● *Related systems already seeing private participation*



ANTI-TANK SYSTEMS
Nag (in pic) anti-tank guided missile (ATGM), SANT (Stand-off Anti-Tank) missile, and man-portable ATGM. SANT is air-to-surface, ideal for helicopters
Range: **4-7 km**
● *Private firm yet to be finalised*

NASM-SR
The air-launched Naval Anti-Ship Missile–Short Range is mainly used by helicopters
Range: **55 km**
● *Private sector manufacturing underway*

ULPGM-V3
The UAV-Launched Precision Guided Missile, tested in May 2026, can be used in air-to-air and air-to-ground modes to engage ground targets and aerial threats
Range: **10 km**
● *Adani Defence involved in production*

VSHORADS
The lightweight, man portable Very Short Range Air Defence System can be used by a soldier to engage low-flying aircraft, drones and helicopters
Range: **8 km** Engagement altitude: **4.5 km**
● *Private firm yet to be finalised*

A private company seeks predictable orders, profit margins and export opportunities, whereas the military's priorities involve reliability, interoperability and the ability to replenish critical systems rapidly during operations. The challenge here is to bring in private capital and manufacturing efficiency without allowing commercial logic to override operational requirements.

SECURITY & SAFEGUARDS

Dr Ravi Gupta, a former DRDO scientist, says decades of patient guidance by the organisation had strengthened the technological abilities of Indian industry to a level where private companies—already involved in manufacturing missile subsystems designed by DRDO—could now absorb whole technologies.

Asked if private players can reliably deliver missiles at scale, Gupta says, "Scale will depend on demand—that is, the quantum and efficiency of supply orders placed by the ministry of defence. No industry will invest in expensive production facilities only to keep them idle because of inadequate orders or bureaucratic delays." However, he flags the risk of monopolies. "The only risk I foresee is the creation of monopolies in the private sector, with a few large industrial houses dominating the arena and killing competition," he says. On additional security challenges related to greater private-sector involvement in sensitive defence production, Gupta says established procedures could mitigate such risks, provided technology transfers were handled carefully. "Proper procedures will have to be put in place to minimise security risks, including both cyber and physical threats," says a former DRDO chief, echoing Gupta.

Indeed, safeguards remain central to the policy. ToTs are granted only to companies that meet strict eligibility criteria, and DRDO retains oversight over quality, intellectual property and future upgrades.

However, successful absorption of complex missile technologies will demand significant private investment in skilled manpower and specialised facilities. India's private defence industry does not yet have the capacity to manufacture sophisticated missile systems at the scale and reliability required by the armed forces, says the former DRDO chief. True, private firms do produce components, but critical capability gaps remain. For instance, they currently lack sufficient capacity for propellant casting for large-diameter rocket motors—essential for missile production. Firms with existing capabilities in propulsion, electronics, seekers and integration—such as Solar Defence & Aerospace, Larsen & Toubro and Bharat Forge—can move to a higher trajectory and invest in advanced manufacturing and testing infrastructure. M.V. Reddy says investments for missile production infrastructure is a significant challenge for private players, as the outlay would be difficult to justify without a sizeable contract from the military.

Another senior DRDO scientist stresses the need for large-scale production. "It is not enough to develop and test a good missile. We must also have the ability to produce it in large numbers and replace stocks during a war." India, he says, needs a strong missile industry where new DRDO technologies could meet the advanced production capacity of both public and private industry. "This is how major military powers have built their strength," he adds.

Though quality control, manufacturing at scale and greater investment are challenges for the private sector for now, the new policy sets the Indian defence industry on the path to develop a robust military-industrial complex essential for self-reliance. Indian missiles made by private defence firms are an essential part of that goal. ■